

EDUCATION**National University of Singapore****Aug 2024 - Sep 2026**

Master of Technology in Artificial Intelligence Systems

- Graduate Diploma in Systems Analysis (Completed during program)
- Relevant course work: Intelligent Sensing Systems, Practical Language Processing, Intelligent Reasoning Systems, Pattern Recognition Systems.

Xiamen University**Sep 2020 - Jun 2024**

Digital Media Technology Bachelor School of Computer Technology

- Relevant course work: Computer Graphics, Human-Computer Interaction, Data Structures and Algorithms

TECHNICAL SKILLS

- Multimodal & Generative AI: Diffusion Models, Vision-Language Models, ControlNet, Representation Learning
- Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, Model Training & Evaluation
- LLM/NLP: RAG, LoRA/PEFT, LangChain, OpenAI API

WORK EXPERIENCE**Software Engineer Intern****Mar 2025 - Aug 2025**

A*STAR-Singapore Institute of Manufacturing Technology

- Developed backend and full-stack modules for a data-intensive sustainability platform, enabling analytical workflows and computation-driven environmental impact assessment.
- Improved LCA computation runtime through sparse matrix optimisation and data caching strategies.
- Built scalable API and data processing components using FastAPI, PostgreSQL, and AWS Lambda for computation-heavy platform functions.

Research Intern (Multimodal GenAI and World Models)**Mar 2026 - Present**

Singapore-MIT Alliance for Research and Technology (SMART)

- Constructed a multimodal generative framework integrating satellite imagery and urban energy data using diffusion-based models and ControlNet conditioning.
- Built and processed large-scale multi-city datasets (MUSE), aligning satellite images with energy consumption labels for cross-modal learning and generation.
- Conducted downstream experiments to evaluate synthetic data effectiveness, demonstrating enhanced segmentation performance (mIoU) under limited real-data regimes.
- Contributed to ongoing research targeting top-tier publication, including model design, ablation studies, and experimental validation of generative data augmentation strategies.

PROJECTS**Multimodal Emotion Recognition System****Aug 2025 - Feb 2026**

Audio-Visual Emotion Recognition with Cross-Modal Fusion

- Integrated ResNet18, WavLM, and cross-attention fusion for joint audio-visual representation learning.
- Conducted comparative experiments across late fusion, concat fusion, gated fusion, and cross-attention fusion.
- Improved model robustness and training stability through two-stage fine-tuning and noise-aware training.

Smart Replenishment for Store Managers**Aug 2025 - Feb 2026**

Explainable AI Workflow for Inventory Decision Support

- Developed an AI decision pipeline combining ARIMA and LSTM demand forecasting with Chain-of-Thought (CoT) reasoning for replenishment recommendations.
- Automated workflow orchestration and API triggers with n8n to streamline data flow and support end-to-end inventory decisions.
- Established an interactive prototype to visualise forecast outputs, reasoning steps, and replenishment decisions in an explainable workflow.